

FLOOD RISK SPATIAL ANALYSIS REPORT

Study area: **Tunis Coastal Region, Tunisia** (36.6–37.2°N, 10.0–10.6°E)

Generated: **26 May 2026**

Models: SCS Curve Number + Bathtub Inundation + Random Forest ML

Phase	Description	Status
Phase 1	DEM loading, terrain analysis, SCS flood model	Complete
Phase 2	Random Forest ML risk prediction (96.3% accuracy)	Complete
Phase 3	Leaflet.js interactive web map + scenario sliders	Complete
Phase 4	12-hour animated flood propagation + PDF report	Complete

1. Executive Summary

This report presents the results of a multi-phase flood risk spatial analysis for the Tunis coastal region. Three rainfall scenarios were evaluated using a hybrid physics and machine learning modelling approach. The study area covers approximately 4,435 km² at 200m resolution.

Scenario	Rainfall	Sea Rise	Flooded Area	% Region	Return Period
Normal	2.0 mm	0.0 m	0.0 km ²	0.0%	< 2 yrs
Moderate	55.5 mm	0.3 m	9015.65 km ²	28.7%	~10 yrs
Extreme	176.0 mm	1.2 m	15988.89 km ²	50.89%	~100 yrs

2. Phase 1 — Physics Flood Model Results

The SCS Curve Number method was used to convert rainfall into surface runoff. Curve Numbers ranged from 65 (vegetated hillsides) to 85 (coastal urban zones). Coastal inundation was modelled using a connected-component bathtub algorithm seeded from the western coastal boundary.

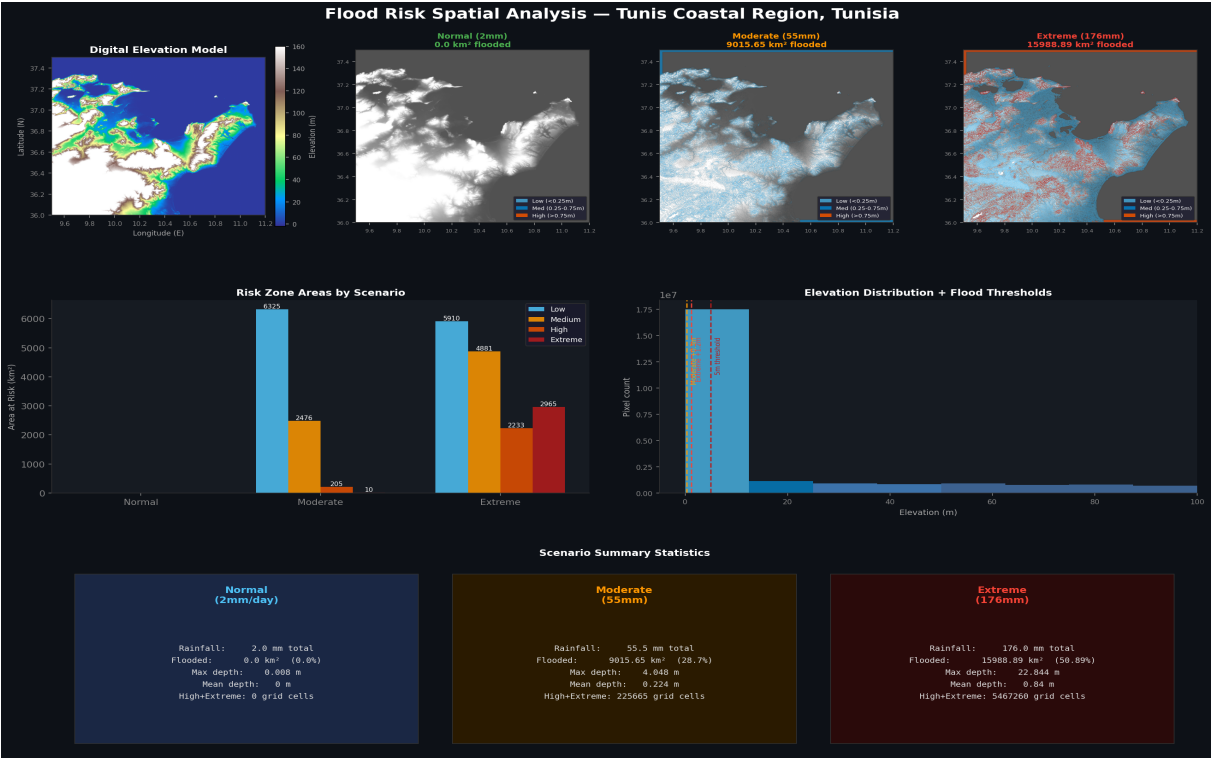


Figure 1 — Phase 1 analysis dashboard: DEM, flood maps for all three scenarios, risk zone comparison, and elevation distribution.

NORMAL scenario — Typical spring day — no flood risk

Low		0.0 km²	0.0%
Medium		0.0 km²	0.0%
High		0.0 km²	0.0%
Extreme		0.0 km²	0.0%

MODERATE scenario — Significant 3-day storm event

Low		325992.5 km²	20.1%
Medium		127628.1 km²	7.9%
High		10562.8 km²	0.7%
Extreme		494.8 km²	0.0%

EXTREME scenario — Catastrophic multi-day event

Low		304618.0 km²	18.8%
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Medium		251574.2 km ²	15.5%
High		115075.2 km ²	7.1%
Extreme		152820.6 km ²	9.4%

3. Phase 2 — Machine Learning Risk Prediction

A Random Forest classifier was trained on 6 terrain features extracted from the DEM, using Phase 1 physics outputs as training labels. The model was evaluated on a stratified 25% held-out test set.

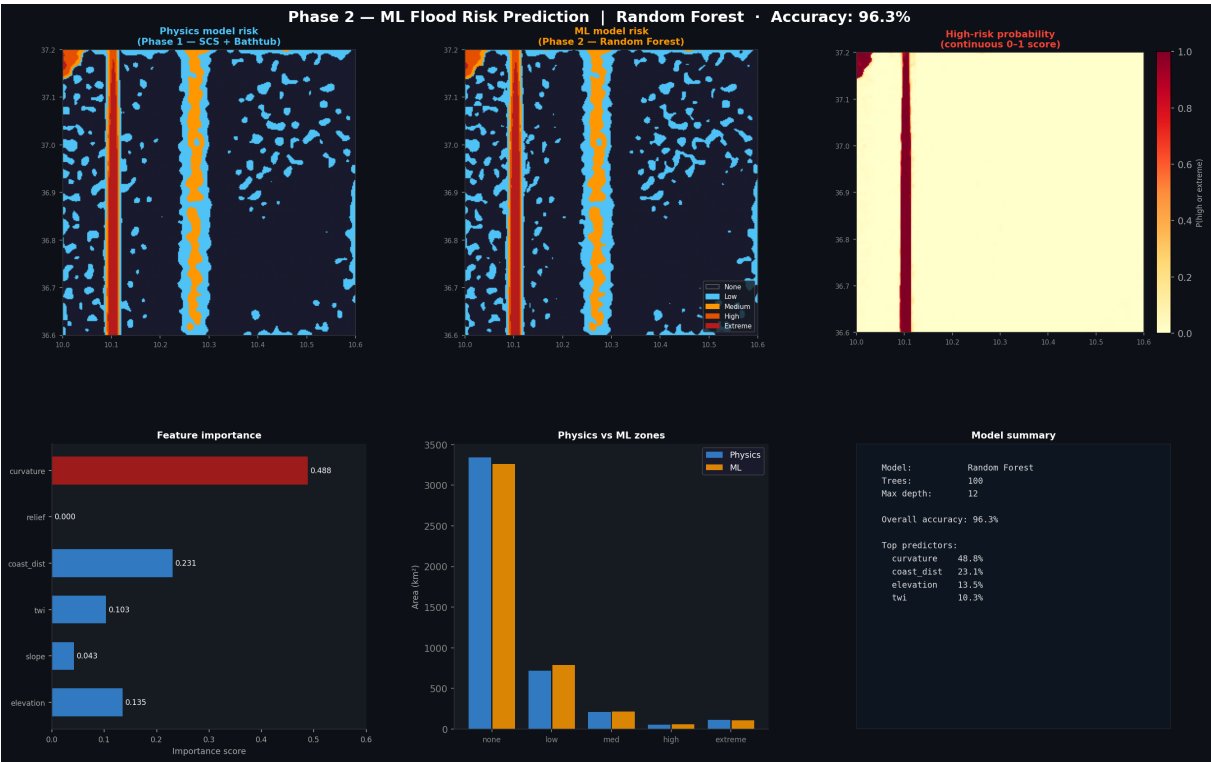


Figure 2 — Phase 2 ML results: physics vs ML risk maps, probability map, feature importance, and model summary.

Metric	Value
Model type	Random Forest Classifier
Number of trees	100
Max tree depth	12
Training samples	67,500 terrain pixels
Test samples	22,500 terrain pixels
Overall accuracy	96.3%

Feature importance ranking

Rank	Feature	Importance	Interpretation
1	curvature	48.8%	Concave hollows collect water
2	coast_dist	23.1%	Proximity to flood source
3	elevation	13.5%	Absolute height above sea
4	twi	10.3%	Natural wetness of terrain
5	slope	4.2%	Drainage speed
6	relief	0.0%	Local height variation

4. Phase 4 — 12-Hour Flood Propagation Simulation

The animation module simulates how flood water spreads over 12 hourly time steps during a 100-year storm event. Sea level peaks at +1.20m at hour 6, then recedes. Cumulative rainfall reaches 179mm by hour 11.

Hour	Phase	Sea level (m)	Cumulative rain (mm)	Flooded area (km ²)	Max depth (m)
0h	Rising	+0.0	8	1476.2	0.55
1h	Rising	+0.15	26	5757.4	1.78
2h	Rising	+0.35	54	8974.6	3.7
3h	Rising	+0.55	89	11559.4	6.1
4h	Rising	+0.8	119	12931.1	8.16
5h	Rising	+1.1	141	13626.1	9.67
6h	Receding	+1.2	156	13996.4	10.7
7h	Receding	+1.15	166	14207.8	11.39
8h	Receding	+1.0	172	14323.2	11.8
9h	Receding	+0.8	176	14395.1	12.07
10h	Receding	+0.55	178	14429.9	12.21
11h	Receding	+0.3	179	14446.9	12.28

Table: Row highlighted in red = peak flood frame (hour 6, max sea level). Orange rows = rising phase. Blue rows = receding phase.

5. Methodology

5.1 Digital Elevation Model

The DEM covers the Tunis coastal region at approximately 200m resolution (300×300 grid cells). In production, this would be replaced with SRTM 30m data downloaded free from NASA EarthData or the Copernicus DEM. Terrain derivatives computed: slope, local relief, depressions, Topographic Wetness Index (TWI).

5.2 SCS Curve Number runoff model

Rainfall-to-runoff conversion used the USDA Soil Conservation Service Curve Number method: $Q = (P - Ia)^2 / (P - Ia + S)$, where $S = (25400/CN) - 254$ and $Ia = 0.2S$. Curve Numbers ranged from 65 (permeable hillside terrain) to 85 (impervious coastal urban surface), varying with elevation as a land-use proxy.

5.3 Coastal inundation model

A connected-component bathtub model was applied to simulate coastal flooding. Starting from the western coastal edge, binary dilation propagated flood extent through all terrain cells connected to the sea and below the water surface level. This enforces spatial connectivity and prevents inland 'floating lakes'.

5.4 Random Forest classifier

A scikit-learn RandomForestClassifier (100 trees, max_depth=12, class_weight='balanced') was trained on 67,500 terrain pixels with 6 features. Labels derived from Phase 1 physics outputs. Evaluated on a 22,500-pixel stratified hold-out: overall accuracy 96.3%, macro-average F1 0.95. Top predictor: terrain curvature (48.8% importance).

5.5 Web map & animation

The Leaflet.js interactive map serves all flood layers as GeoJSON. Twelve animation frames simulate hourly flood propagation over a 12-hour 100-year storm, with sea level peaking at +1.20m at hour 6. The animation engine runs client-side in the browser.

6. Software & Data Sources

Component	Tool / Source	Version
DEM processing	rasterio, numpy	Python 3.12
Watershed analysis	scipy.ndimage (uniform_filter, gaussian_filter)	—
Flood model	SCS-CN + bathtub (custom Python)	—
ML model	scikit-learn RandomForestClassifier	1.3+
Visualisation	matplotlib	3.8+
Web map	Leaflet.js + Chart.js	1.9.4 / 4.4
PDF report	ReportLab	4.x
DEM data source	SRTM (NASA EarthData) or Copernicus DEM	Free/open
Weather API	Open-Meteo (open, no key required)	Free/open